Table 2. Linear regression models testing genetic and sex-specific effects on immune-predicted weigh

Progressive model building from simple genetic effects to complex interactions

Model	Predictors	β[1]	Model Fit	Interpretation
Model 1: Genetic baseline			R^2 = 0.019	No simple genetic effect
	HI	-0.15 (0.67)		
	He	-0.39 (2.04)		
	HI × He	4.02 (3.44)	AIC = 1419	
Model 2: Main effects			R^2 = 0.063	Infection dominates
	Sex (Male)	-0.09 (0.28)		
	HI	0.19 (0.41)		
	He	1.56 (1.01)		
	Infection	**1.12 (0.29)*** **	AIC = 1407	
Model 3: Two-way interactions			R^2 = 0.093	Sex-specific genetic effe
	Sex (Male)	-1.31 (0.74)		
	HI	-0.67 (0.83)		
	Infection	2.32 (0.80)**		
	Sex × HI	**2.14 (0.83)***		Males worsen with musc
	HI × Infection	**-1.30 (0.84)**	AIC = 1409	
Model 4: Full factorial			R^2 = 0.112	Overparameterized
	[15 parameters]	—	AIC = 1413	
Model 5: Focused			R^2 = 0.099	Three-way interaction N
	[12 parameters]	—	AIC = 1411	
Model 6: Species-specific			R^2 = 0.147[2]	Species matters but
	E. falciformis	**5.22 (1.75)** **		
	E. ferrisi	1.43 (1.07)	n = 168	reduced sample size

[1] Key findings highlighted in bold. Significance: ***p < 0.001, **p < 0.01, *p < 0.05

[2] AIC not comparable due to different sample size

Model Selection: Best fit = Model 2 (lowest AIC); Most informative = Model 3 (biological insight); SE = standard error; NS = not significant